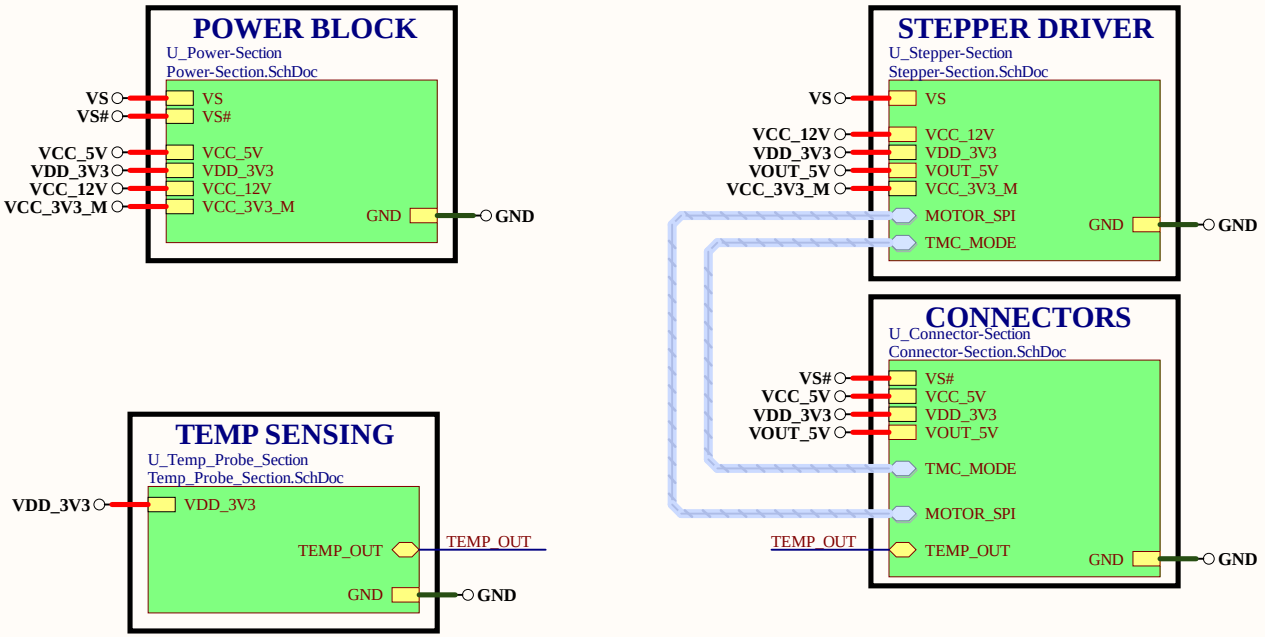
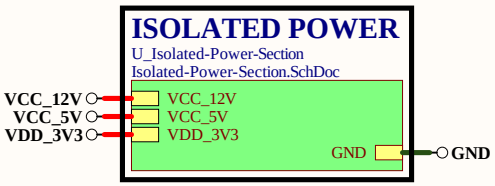
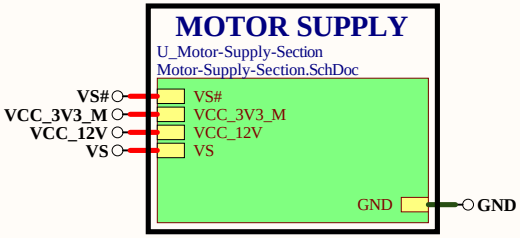
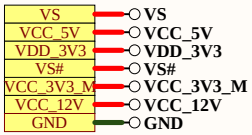
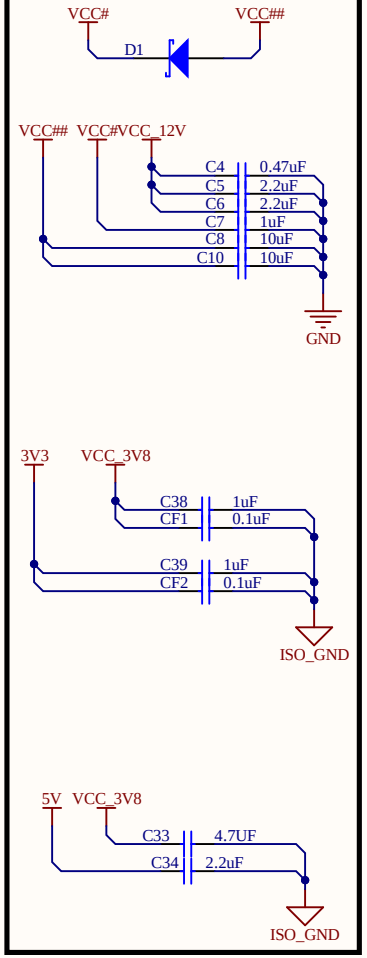

POWER BLOCK
U_Power-Section
Power-Section.SchDoc
VS
VS#
VCC_5V
VDD_3V3
VCC_12V
VCC_3V3_M
GND
GND
STEPPER DRIVER
U_Stepper-Section
Stepper-Section.SchDoc
VS
VCC_12V
VDD_3V3
VOUT_5V
VCC_3V3_M
MOTOR_SPI
TMC_MODE
GND
GND
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc
VS#
VCC_5V
VDD_3V3
VOUT_5V
TMC_MODE
MOTOR_SPI
TEMP_OUT
GND
GND
TEMP SENSING
U_Temp_Probe_Section
Temp_Probe_Section.SchDoc
VDD_3V3
TEMP_OUT
GND
GND
TEMP_OUT


ISOLATED POWER
U_Isolated-Power-Section
Isolated-Power-Section.SchDoc
VCC_12V
VCC_5V
VDD_3V3
GND
VCC_12V
VCC_5V
VDD_3V3
GND
GND

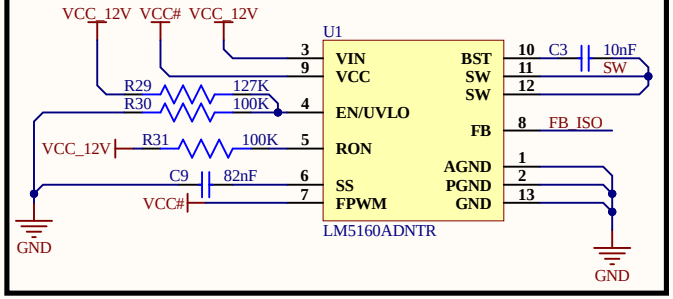

MOTOR SUPPLY
U_Motor-Supply-Section
Motor-Supply-Section.SchDoc
VS#
VCC_3V3_M
VCC_12V
VS
VS#
VCC_3V3_M
VCC_12V
VS
GND
GND


VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND
VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND

DECOUPLING CAPS

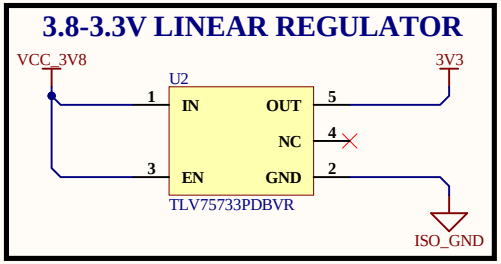


12V-3.8V ISOLATED BUCK CONVERTER

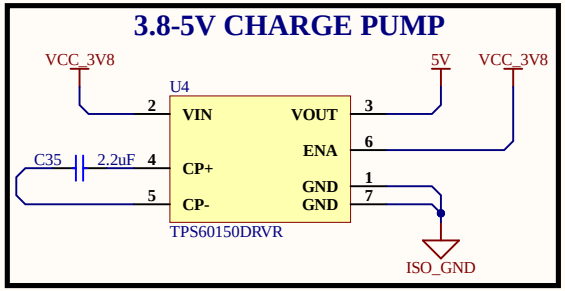


Primary side output voltage would be 5.7 for a 3.8V VSec (NP/Ns=Vpri/Vsec)
FB resistors RF2=3.7K and RF1=2K.
Max current of this transformer is 840mA. (suggested in datasheet)

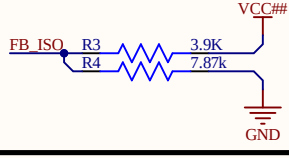
3.8-3.3V LINEAR REGULATOR



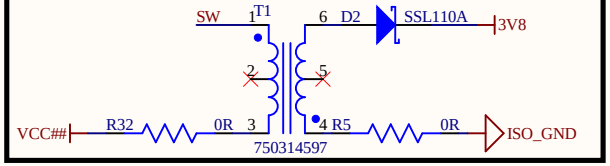
3.8-5V CHARGE PUMP



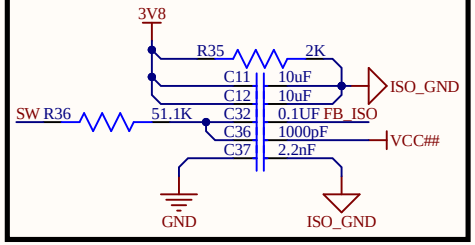
FEEDBACK NETWORK



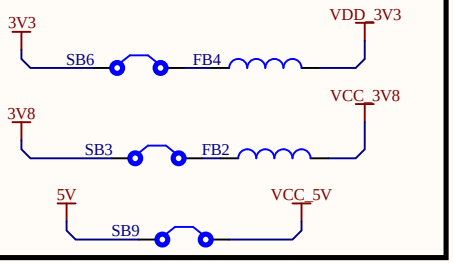
FLYBACK TRANSFORMER



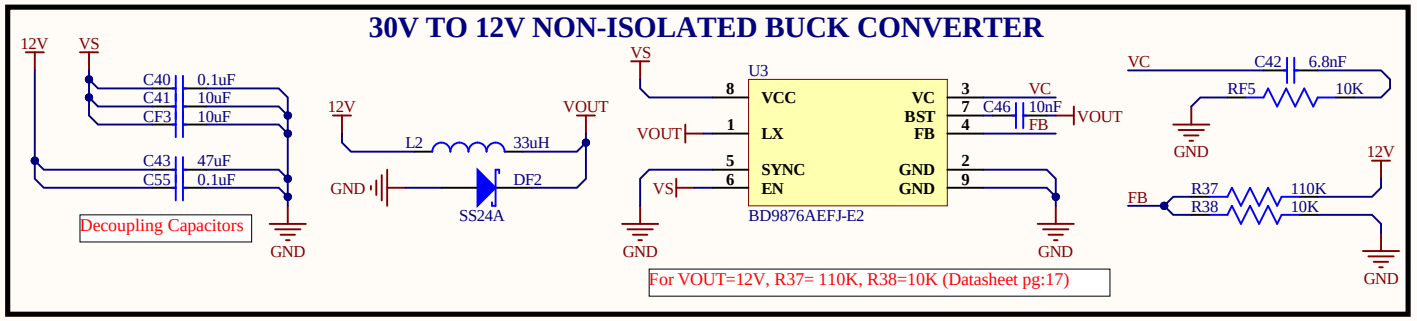
SUPPORTING CIRCUIT

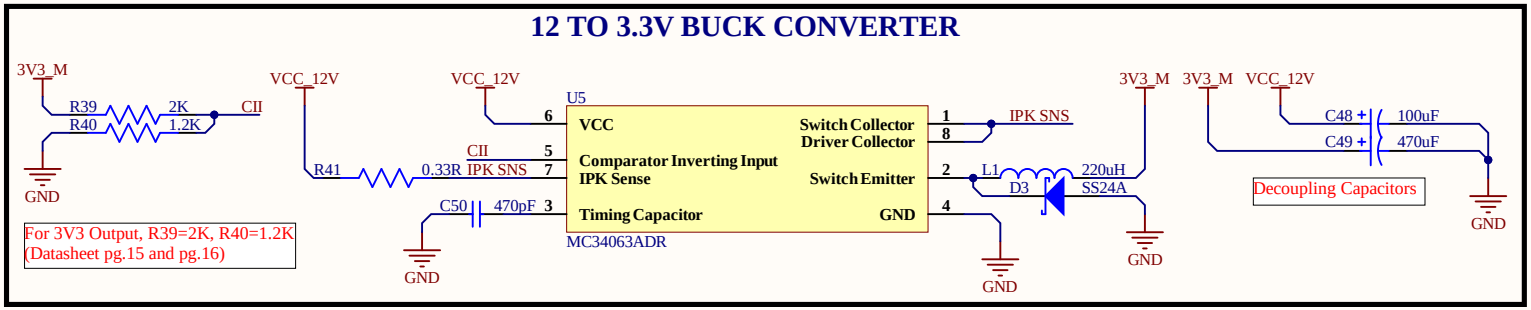


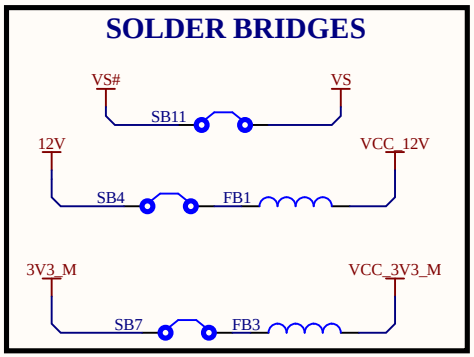
SOLDER BRIDGE CONNECTIONS

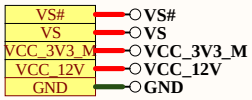


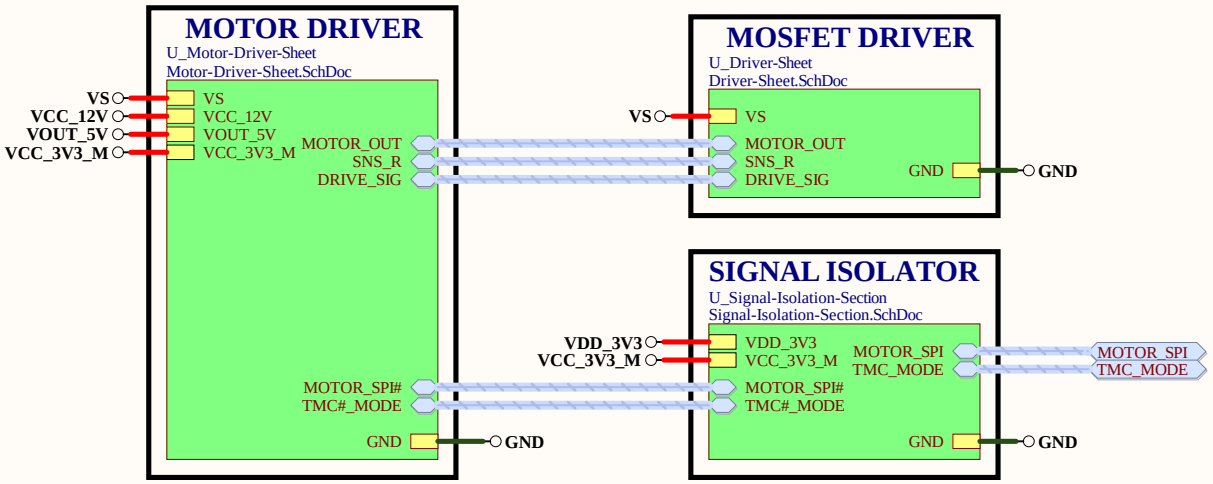
VCC_12V — VCC_12V
VCC_5V — VCC_5V
VDD_3V3 — VDD_3V3
GND — GND

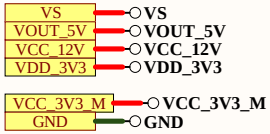

30V TO 12V NON-ISOLATED BUCK CONVERTER
12V
VS
C40 0.1uF
C41 10uF
CF3 10uF
C43 47uF
C55 0.1uF
Decoupling Capacitors
GND
12V
VOUT
L2 33uH
GND
DF2
SS24A
VS
U3
8 VCC
1 LX
5 SYNC
6 EN
VC 3
BST 7
FB 4
GND 2
GND 9
C46 10nF
VOUT
VOUT
VS
GND
GND
BD9876AEFJ-E2
VC
C42 6.8nF
RF5 10K
GND
12V
FB
R37 110K
R38 10K
GND
For VOUT=12V, R37= 110K, R38=10K (Datasheet pg:17)

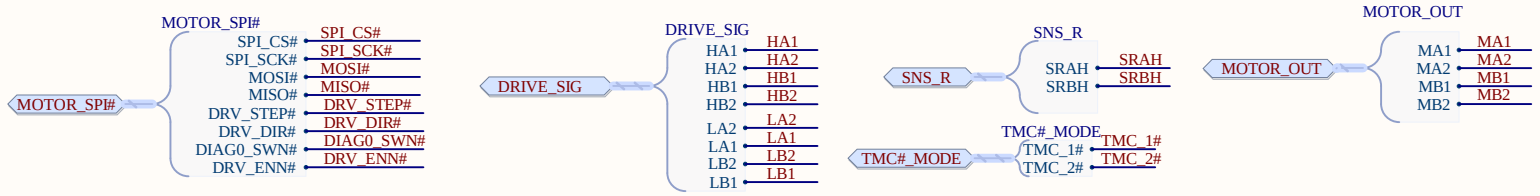

12 TO 3.3V BUCK CONVERTER
3V3_M
R39 2K
R40 1.2K
CII
GND
For 3V3 Output, R39=2K, R40=1.2K
(Datasheet pg.15 and pg.16)
VCC_12V
R41 0.33R
CII
IPK SNS
C50 470pF
GND
VCC_12V
U5
6 VCC
5 Comparator Inverting Input
7 IPK Sense
3 Timing Capacitor
Switch Collector 1
Driver Collector 8
Switch Emitter 2
GND 4
MC34063ADR
IPK SNS
L1 220uH
D3 SS24A
GND
GND
3V3_M
3V3_M
VCC_12V
C48 100uF
C49 470uF
Decoupling Capacitors
GND


SOLDER BRIDGES
VS#
VS
SB11
12V
VCC_12V
SB4
FB1
3V3_M
VCC_3V3_M
SB7
FB3

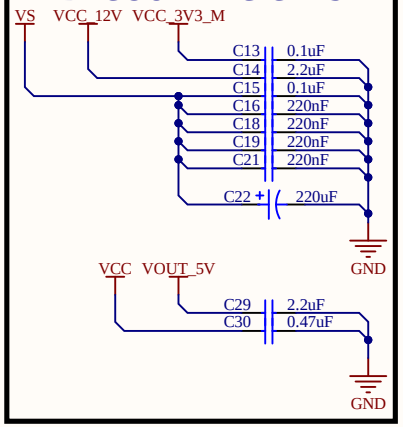

VS#
VS
VCC_3V3_M
VCC_12V
GND
VS#
VS
VCC_3V3_M
VCC_12V
GND


MOTOR DRIVER
U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
VCC_12V
VOUT_5V
VCC_3V3_M
MOTOR_OUT
SNS_R
DRIVE_SIG
MOTOR_SPI#
TMC#_MODE
GND
MOSFET DRIVER
U_Driver-Sheet
Driver-Sheet.SchDoc
SIGNAL ISOLATOR
U_Signal-Isolation-Section
Signal-Isolation-Section.SchDoc
VDD_3V3
MOTOR_SPI
TMC_MODE


VS
VOUT_5V
VCC_12V
VDD_3V3
VCC_3V3_M
GND



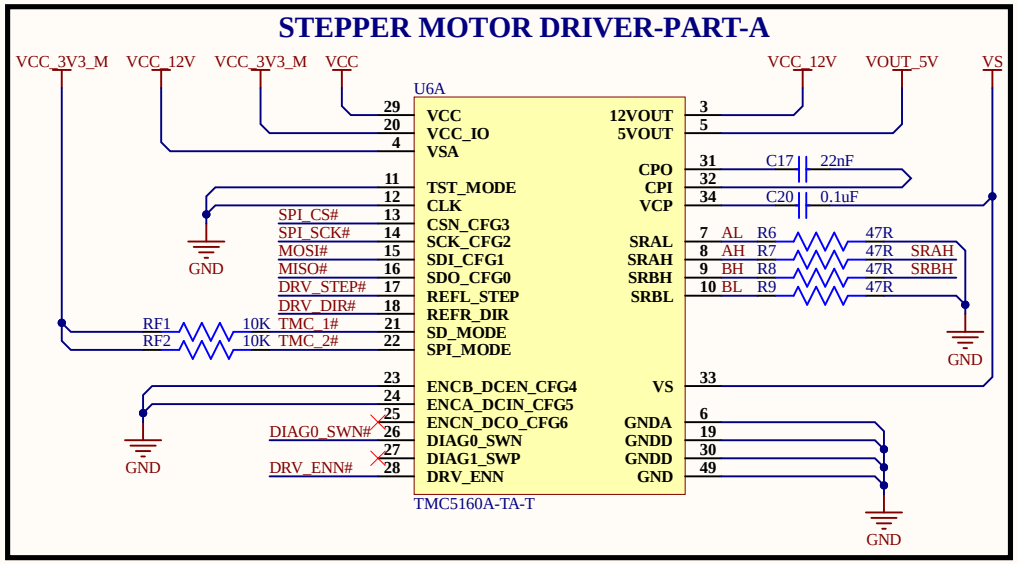
DECOUPLING CAPS



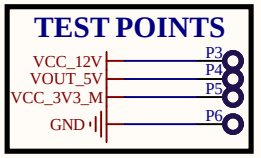
SERIES RESISTOR



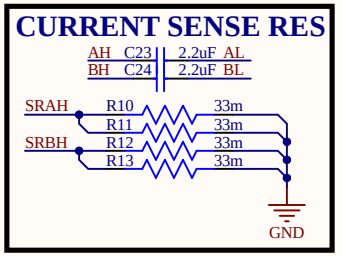
STEPPER MOTOR DRIVER-PART-A



TEST POINTS



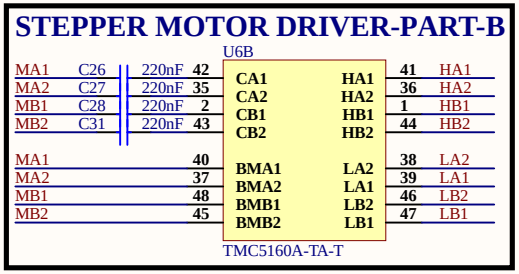
CURRENT SENSE RES



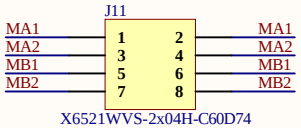
MOTOR OUTPUT

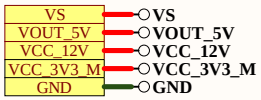


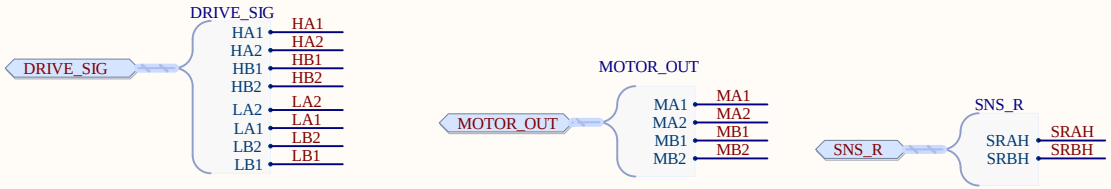
STEPPER MOTOR DRIVER-PART-B



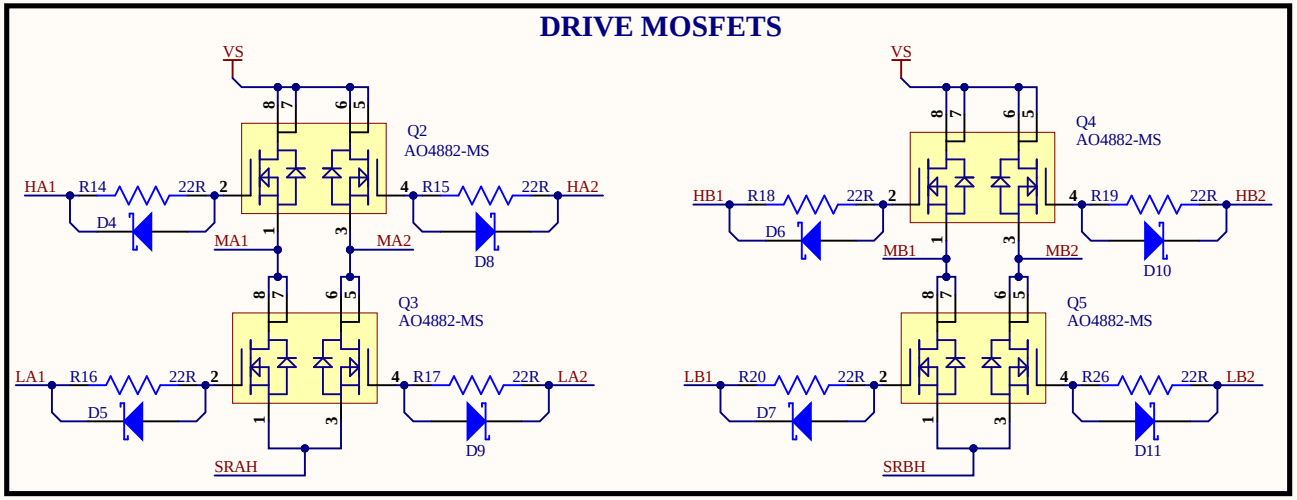
MOTOR OUTPUT FOR NEMA34



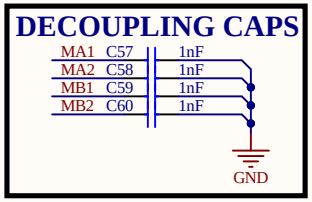




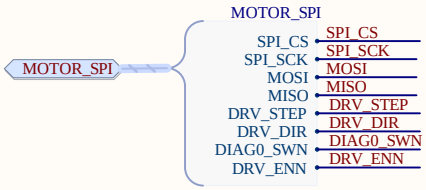
DRIVE MOSFETS

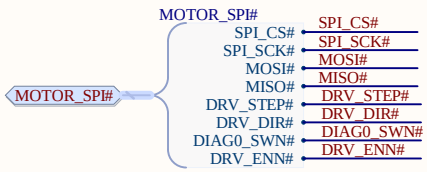


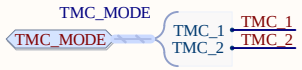
DECOUPLING CAPS



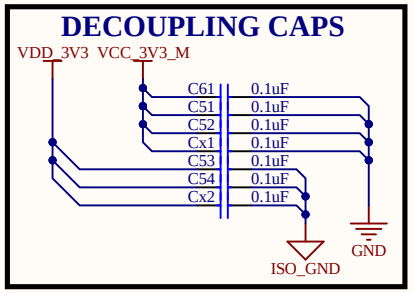


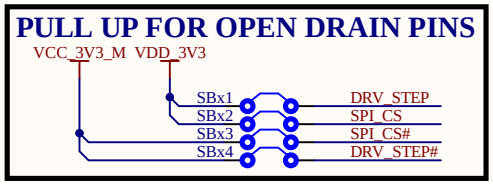




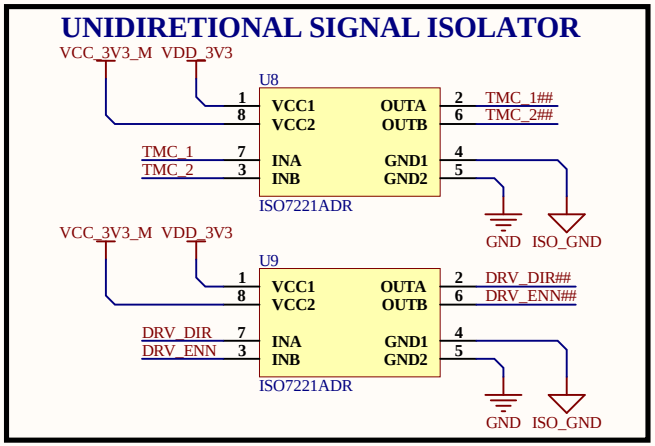


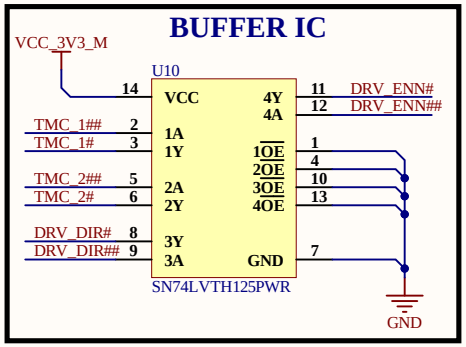


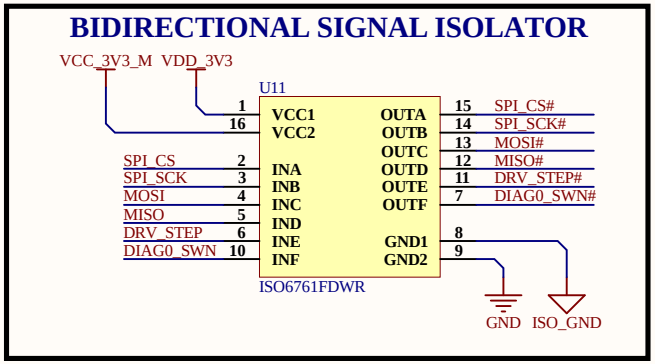


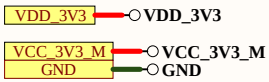


Mount 1K 0603 resistors on these bridges if needed





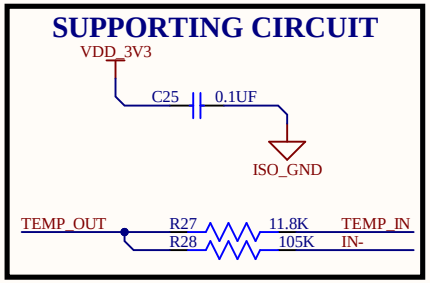




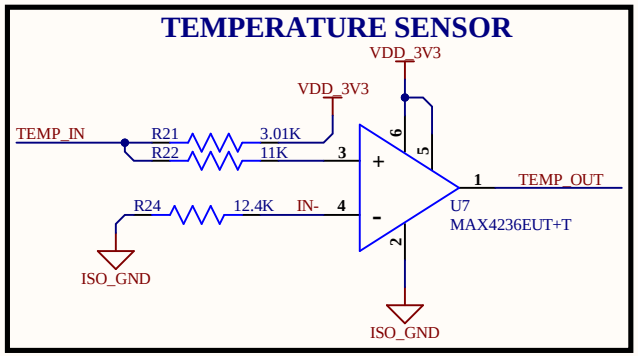
Title		
Size A4	Number	Revision
Date: 11/11/2022	Sheet of	
File: C:\GitHub\..\Signal-Isolation-Section.SchDoc	Drawn By:	

TEMP_OUT TEMP_OUT

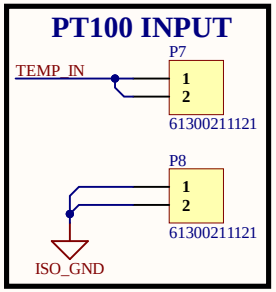
SUPPORTING CIRCUIT



TEMPERATURE SENSOR



PT100 INPUT



VDD_3V3 **VDD_3V3**
GND **GND**

INDICATION FOR MOTOR 5V SUPPLY

TMC_MODE
TMC_MODE
TMC_1
TMC_2
TMC_1
TMC_2

TEMP_OUT
TEMP_OUT

VOUT_5V
R25
143R
LED2
GND

3.3V RGB LED

VDD_3V3
LED1
R33 51R 6 1 LED_B
R34 150R 5 2 LED_R
R46 51R 4 3 LED_G
LTST-G683GEBW

13 MM ACTUATOR

SW1
BUTTON
3
4
C2
0.1uF
1
2
Tactile Switch 13mm Actuator
ISO_GND

HEADERS TO SELECT BOOT MODE

VDD_3V3
RF3 10K BOOT1
RF4 10K BOOT0
Default: Embedded SRAM Mode

P9
1
2
BOOT1
61300211121
ISO_GND

P10
1
2
BOOT0
61300211121
ISO_GND

GND BOOT1 to enter
System Memory Bootloader Mode

GND BOOT0 to enter
User Flash Memory Mode

MOTOR_SPI
MOTOR_SPI
SPI_CS SPI_CS
SPI_SCK SPI_SCK
MOSI MOSI
MISO MISO
DRV_STEP DRV_STEP
DRV_DIR DRV_DIR
DIAG0_SWN DIAG0_SWN
DRV_ENN DRV_ENN

USB TYPE C FOR DATA TRANSFER

J1
A4 A9 B4 B9 VBUS
A8 B8 RFU1 RFU1
A2 B11 SSTXp1
A3 B10 SSRXn1
USB_DP A6 Dp1
USB_DM B7 Dn1
USB_DM A7 Dn1
USB_DP B6 Dp1
CC1 A5
CC1 B5
R1 5.1K
R2 5.1K
ISO_GND
SSRXn2 A10
SSTXn2 B3
SSRXn2 A11
SSTXp2 B2
GND A1
GND B1
GND A12
GND B12
MH1
MH2
MP1
MP2
MP3
MP4
MC-318FB
ISO_GND

BOARD TO BOARD CONNECTOR

J3
VCC_5V
VDD_3V3
LED_R A1 B1 LED_G
USB_DM A2 B2 SPI_CS
USB_DP A3 B3 SPI_SCK
CAN_NL A4 B4 MOSI
CAN_NH A5 B5 MISO
BOOT0 A6 B6 DRV_STEP
LED_B A7 B7 DRV_DIR
BUTTON A8 B8 DIAG0_SWN
TMC_2 A9 B9 DRV_ENN
TMC_1 A10 B10 TEMP_OUT
BOOT1 A11 B11
A12 B12
A13 B13
MP1 MP2
MP3 MP4
MH1 MH2
M55-7012642R
ISO_GND
ISO_GND

DECOUPLING CAP

VS#
C1
220uF
GND

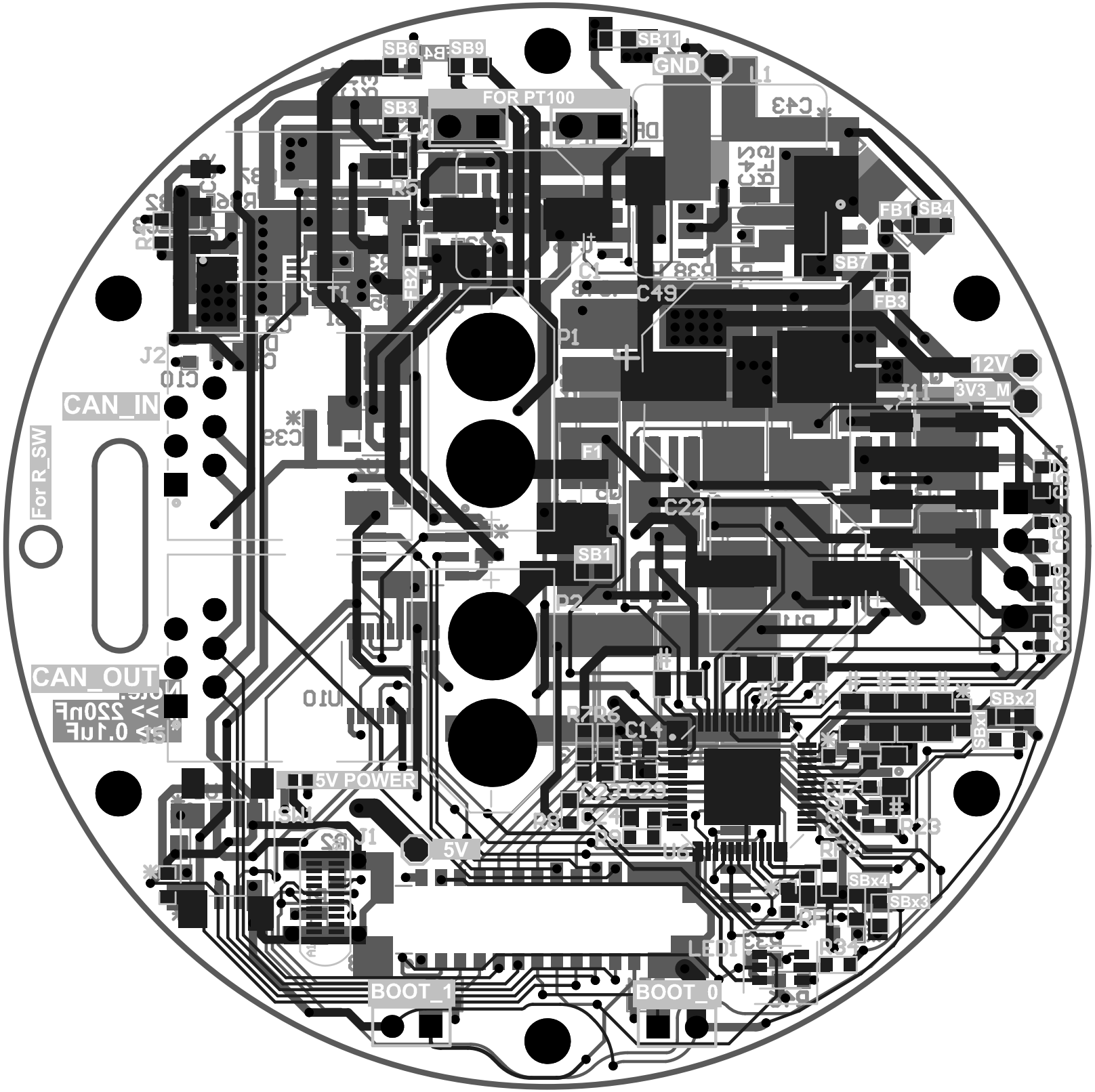
INPUT CONNECTORS

VS#
F1
P1
2
1
PRT-10474
GND
VS#
SB1
P2
2
1
PRT-10474
GND

RJ11 FOR CAN BUS

J2
CAN_NL
CAN_NH
1 2 3 4 5 6 MH1 MH2
5520425-3
CAN IN
J5
CAN_NH
CAN_NL
1 2 3 4 5 6 MH1 MH2
5520425-3
CAN OUT

VS# VS#
VCC_5V VCC_5V
VDD_3V3 VDD_3V3
VOUT_5V VOUT_5V
GND GND


SB6
SB9
SB11
GND
FOR PT100
SB3
FB1
SB4
SB7
FB3
C49
P1
12V
3V3_M
CAN_IN
For R_SW
F1
C22
SB1
P2
CAN_OUT
SBx2
5V POWER
5V
BOOT_1
BOOT_0